Introduction to Big Data
Chapter 1
Clustering (CL)

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Introduction

Applications and data

Applications

Segmentation of customers files, used in the banking and insurance sectors but also by many other companies especially for Market studies.

The data

There exist several cluster analysis methods depending on the type of data. In what follows we focus on tables of the form units (rows) by variables (columns).

More precisely, we assume that we have n observations described by p quantitative variables which form a table of size $n \times p$.

Important remark: when the variables are very heterogeneous (different variances or scales), standardizing the data is highly recommended

Notations : $X_1, \ldots, X_n$ denote the n individuals and $X^1, X^2, \ldots, X^p$ the p numerical variables.

In marketing, the aim is to exhibit groups (clusters) of customers whose socio-economic characteristics and purchasing behavior are homogeneous. The quantitative variables may be the age, the sex and the quantities or amounts of the products purchased.

Objectives

Cluster analysis (CA) is a multivariate data analysis technique allowing to build groups of individuals such that:

- each group is homogeneous according to some characteristics, that is the observations of a group are as alike as possible,
- each group is different from the others according to the same characteristics, that is the observations from a group differ as much as possible from those of the other groups.

With a clustering method, each observation is classified into a given group.

Introduction

Similarity measure and inertia

Similarity measure

The concepts of similarity or difference between individuals are formalized by measuring **distances** between individuals. In the following, we will work with the usual euclidian distance:

$$d(X_i, X_j) = \sqrt{\sum_{k=1}^{p} (X_i^k - X_j^k)^2}.$$

Inertia

In clustering, we also consider inertia which is linked with the distance between observations and the mean vector $\overline{X} = (\overline{X^1}, \overline{X^2}, \cdots, \overline{X^p})$:

$$I = \sum_{j=1}^{p} Var(X^j) = \frac{1}{n} \sum_{j=1}^{p} \sum_{i=1}^{n} (X_i^j - \overline{X^j})^2 = \frac{1}{n} \sum_{i=1}^{n} d^2(X_i, \overline{X})$$

Introduction

Objectives in terms of inertia

Let us consider a partition in k groups C_1, $C_2, \cdots C_k$, with respective frequencies $n_1, n_2, \cdots, n_k$ $\left(\sum_{r=1}^{k} n_r = n\right)$.

Reminder: a **partition** of a set of observations $\{X_1, \ldots, X_n\}$ is a set of k groups $A_1, \ldots, A_k$ such that :

- $A_1 \bigcup A_2 \ldots \bigcup A_k = \{X_1, \ldots, X_n\}$,
- $\forall i, j \in \{1, \ldots, k\}, i \neq j, A_i \bigcap A_j = \emptyset$

The total inertia I can be written in the following way:

$$I = \sum_{r=1}^{k} \sum_{i \in C_r} \frac{1}{n} d^2(X_i, \overline{X})$$

Moreover, the total inertia I can be decomposed in:

Proposition

$$I = \textit{Within-cluster Inertia} + \textit{Between-cluster Inertia}$$

where

$$\textbf{Within-cluster Inertia} = \sum_{r=1}^{k} \frac{1}{n} \sum_{i \in C_r} d^2(X_i, \overline{X}^{(r)}) = \sum_{r=1}^{k} \frac{n_r}{n} I_r$$

with $I_r = \dfrac{1}{n_r} \displaystyle\sum_{i \in C_r} d^2(X_i, \overline{X}^{(r)})$ the inertia for the cluster C_r

$$\textbf{Between-cluster Inertia} = \sum_{r=1}^{k} \frac{n_r}{n} d^2(\overline{X}^{(r)}, \overline{X})$$

with $\overline{X}^{(r)}$ mean vector of the cluster C_r.

General points

Objectives in terms of inertia

In order to fulfill the objectives, clustering methods aim at finding clusters such that the between-cluster inertia is maximized while the within-cluster inertia is minimized.

Presentation of the clustering techniques

Clustering methods are based on some **algorithms**. It means that the clusters are defined through an algorithm that consists in a series of operations defined in a repetitive manner.

There exist two main families of clustering methods: the **partitioning** ones and the **hierarchical** ones.

A classification by direct partitioning consists in directly searching a partition of the individuals.

The obtained groups aim at maximizing the between inertia and at minimizing the within inertia.

There exist different methods for partitioning. The k-means method is the most popular.

Drawback: the number of cluster has to be fixed in advance (it is usually unknown).

Advantage: this method is very fast even for very big tables.

Presentation of the algorithm

- The method first consists in fixing *a priori* a number k of clusters and in representing each group by a particular individual from the data set.
- Every remaining individual is allocated to its closest group.
- The centers (or means) of each cluster are computed and the individuals are possibly reallocated to the closest group.
- This last step is repeated (iterations) until the centers of the clusters are little or not modified.

From a theoretical point of view, it can be shown that the previous algorithm converges under general assumptions. Moreover the within variance decreases at each iteration.

This algorithm has numerous variants according to :

- the way of choosing the individuals which represents the clusters at the beginning (the initial “centers”),
- the frequency for computing the centers (after the allocation of each individual or after all the allocations),
- the stopping criterion of the procedure.

Ascending hierarchical clustering (AHC)

Principles

Principle

The method consists in considering as an initial typology as many clusters as individuals and in grouping together the clusters by successive steps until obtaining only one cluster containing all the individuals.

Ascending hierarchical clustering (AHC)

Algorithm description

Algorithm

Initialization : all the distances between individuals are computed (table of size $n \times n$). Each individual represents a cluster.

Iterations :

- the 2 closest clusters are aggregated in a new cluster,
- the table of distances is updated, taking into account the new cluster (computing of the distances of this new cluster with the other clusters).

Iterations stop when there is only one cluster.

Ascending hierarchical clustering (AHC)

Distances between clusters

Distances between clusters for the aggregating procedure:

In the presentation of the above algorithm, we use the concept of "distance" between **clusters** and not only observations. This concept must be specified and, according to the chosen distance, different algorithms are obtained.

In the following, we propose to focus on a particular choice which consists in using an euclidian distance between the clusters centers.

If we denote by c_a (respectively c_b) the center of the cluster a of size n_a (respectively b of size n_b), the used distance is the **Ward jump** defined by

$$D^2_{ward}(a, b) = \frac{n_a n_b}{n_a + n_b} d^2(c_a, c_b).$$

The Ward jump is the most used distance in practice. It allows to minimize at each iteration the decreasing of the between-class inertia. As a matter of fact, when two clusters are aggregated, the between-class inertia necessarily decreases but one wishes that it decreases the least possible since the clusters must be separated at best.

There exist other aggregation methods

- the **minimal jump**:$D_{min}(a,b) = \inf\{d(i,j), i \in a, j \in b\}$
- the **diameter** method: $D_{diam}(a,b) = \sup\{d(i,j), i \in a, j \in b\}$
- the **mean distance**: $D_{mean}(a,b) = \frac{1}{n_a n_b} \sum_{i \in a, j \in b} d(i,j)$

Ascending hierarchical clustering (AHC)

The dendrogram
It is a representation, in a tree form, of the successive groupings obtained by AHC.
The height in our context corresponds to the loss in terms of between-inertia when aggregating two clusters. You cut the dendrogram if you consider that you loose too much inertia by taking less clusters.

The interpretation of the groups Interpretation of the means (and standard deviations), scatterplots,. . . ...

Mixed clustering

There exist different ways to combine the kmeans algorithm with hierarchical clustering.

One procedure is as follows:

1. Compute hierarchical clustering and cut the tree into k clusters,
2. Compute the center (i.e the mean) of each cluster,
3. Use the kmeans algorithm with the set of cluster centers (defined in step 2) as the initial cluster centers

In this context, we say that the kmeans algorithm consolidates the AHC.

Mixed clustering

Another procedure for mixed clustering

The advantage of AHC comparatively to the k-means method is that it is not necessary to know the number of clusters in advance.

The dendrogram gives guideline for the choice of the number of clusters.

However, this method, contrarily to the k-means method, is very expensive in computing time and cannot be used if the number of observations is very large (millions). Moreover the dendrogram cannot be drawn as soon as there are thousands of observations

In order to cumulate the advantages of the two methods, the following strategy is proposed :

- let us denote n the number of observations of the data file (for example $n = 10^6$).
- The AHC cannot be used but the kmeans algorithm can be used by choosing a large number k_{FAST} of clusters (for example $k_{\mathrm{FAST}} = 500$).
  The interest in choosing this large number of clusters is that the obtained partition only aggregates individuals that are very similar.
- Then consider the 500 means associated with the 500 clusters and perform a AHC with the 500 centers as observations.
- With 500 observations (which are in reality means), the procedure can now work and we can choose a number k of clusters using the dendrogram.

Conclusion

Clustering analysis is adapted to

- tables of type individuals / variables with quantitative variables,
- when the objective is to define a segmentation of individuals in order to identify target observations (usually customers).
- The interpretation of the clusters is crucial. A clustering is a success if the obtained typology is clear and useful.